

Yassine Chemingui

Research Scientist · Safe Reinforcement Learning · Offline Decision-Making · Black-box Optimization

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RESEARCH SUMMARY

Ph.D. researcher in **offline decision-making**: learning to optimize and act safely from logged data when new experiments are costly or risky. I develop methods that turn static datasets into reliable **control policies** and **optimization models**, with first-author work at **NeurIPS**, **AAAI** (Oral), and **UAI**. My work targets settings where online trial and error is not an option, such as materials discovery, healthcare, and energy systems.

- ▶ **Safe Reinforcement Learning**: policies that maximize reward under safety constraints, adapt to constraints that change at deployment without retraining, and guard against rare but severe failures.
- ▶ **Offline Model-Based Optimization**: surrogate models and search strategies that discover high-performing designs in high-dimensional spaces using only past experimental data.

EDUCATION

Ph.D., Computer Science Outstanding Dissertation Award 2022 – 2027 (expected)
Washington State University Pullman, WA, USA

- ▶ Advisor: [Prof. Janardhan Rao \(Jana\) Doppa](#)
- ▶ Thesis: *Advances in Offline Decision-Making: Black-box Optimization, Safe Reinforcement Learning, and Policy Comparison from Logged Data*

Engineering Degree (Diplôme d'Ingénieur), Signals & Systems Graduated with Excellence 2015 – 2018
Tunisia Polytechnic School Tunis, Tunisia

SELECTED PUBLICATIONS

* denotes equal contribution.

UAI'26 **SteinGate: Tail-Sensitive Safe Reinforcement Learning via Stein Discrepancy**

Yassine Chemingui*, Chenhua Fan*, Honghao Wei, Janardhan Rao Doppa.
Conference on Uncertainty in Artificial Intelligence (UAI), 2026. [\[arXiv\]](#) [\[Code\]](#)

AAAI'26 **Nanoporous Materials Discovery via Search Bias-Guided Surrogate Modeling**

Azza Fadhel, Yassine Chemingui, Minh Hoang, Aryan Deshwal, Trong Nghia Hoang, Janardhan Rao Doppa.
AAAI Conference on Artificial Intelligence (AAAI), 2026.

NeurIPS'25 **O3SRL: Online Optimization for Offline Safe Reinforcement Learning**

Yassine Chemingui, Aryan Deshwal, Alan Fern, Thanh Nguyen-Tang, Janardhan Rao Doppa.
Conference on Neural Information Processing Systems (NeurIPS), 2025. [\[arXiv\]](#) [\[Code\]](#)

AAAI'25 **Constraint-Adaptive Policy Switching for Offline Safe Reinforcement Learning**

Yassine Chemingui, Aryan Deshwal, Honghao Wei, Alan Fern, Janardhan Rao Doppa.
AAAI Conference on Artificial Intelligence (AAAI), 2025. Oral [\[arXiv\]](#) [\[Code\]](#)

AAAI'24 **Offline Model-Based Optimization via Policy-Guided Gradient Search**

Yassine Chemingui, Aryan Deshwal, Trong Nghia Hoang, Janardhan Rao Doppa.
AAAI Conference on Artificial Intelligence (AAAI), 2024. [\[arXiv\]](#) [\[Code\]](#)

Earlier work: RL-based energy management (*Energies*, 2020); deep learning for appliance identification (EECSS, 2021); policy-gradient RL for supply chains (ICASET, 2020).

RESEARCH EXPERIENCE

Graduate Research Assistant

School of EECS, Washington State University

2022 – Present

Pullman, WA, USA

- ▶ Developed offline safe RL algorithms that learn constraint-satisfying policies purely from logged data: a minimax formulation with bandit-style updates for fixed cost budgets (NeurIPS'25), and policy switching that adapts to new safety thresholds at deployment without retraining (AAAI'25 Oral).
- ▶ Introduced tail-sensitive safe RL via Stein discrepancy to guard against rare but severe constraint violations (UAI'26).
- ▶ Recast offline black-box optimization as an RL problem to learn search policies over surrogate models (AAAI'24), and designed search-bias-guided surrogates for data-driven nanoporous materials discovery (AAAI'26).
- ▶ Released open-source implementations of all first-author methods.

Research Assistant

Department of Electrical Engineering, Qatar University

2019 – 2021

Doha, Qatar

- ▶ Built a reinforcement learning-based energy management system for school buildings (*Energies*, 2020).
- ▶ Developed a deep learning model to identify electric appliances from their switching-on current waveforms.

Machine Learning Fellow

Fellowship.AI

2021

USA (Remote)

- ▶ Automated domain-specific chatbot creation by integrating the RASA framework with Facebook's BlenderBot.

Applied Mathematics Engineer

ADAGOS

2018 – 2019

Tunis, Tunisia

- ▶ Developed machine learning solutions on the company's neural network tools and contributed to internal research.

SELECTED AWARDS & HONORS

Outstanding Dissertation Award, School of EECS, Washington State University

2026

Oral Presentation, AAAI Conference on Artificial Intelligence

2025

Outstanding Research Assistant Awards (Computer Science; School of EECS), WSU

2025

Edmund O. Schweitzer III Scholarship, School of EECS, WSU

2025 – 2026

Mahmoud M. Dillsi Family Graduate Fellowship & Alfred Suksdorf Fellowship, WSU

2023 – 2024

Tunisia National Rank 49 (Top 2%), National Entrance Exam for Engineering Schools

2015

SERVICE

Reviewer: NeurIPS, ICML, ICLR, AAAI, AISTATS (2024 – 2027)

NeurIPS Top Reviewer 2024 ICML Gold Reviewer 2026 ICLR Top 200 Reviewer 2026

Teaching Assistant: CptS 437, Introduction to Machine Learning, WSU (Fall 2023)

TECHNICAL SKILLS

Research areas Offline & safe RL, constrained MDPs, black-box & model-based optimization, surrogate modeling

Tools Python, PyTorch, Jax, TensorFlow, NumPy, Gym/Gymnasium, Weights & Biases, Linux